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Continuous Improvement through Back-office Processes Automatization: a case study on Standardized Work Documentation

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Abstract

Nowadays companies operate in a competitive environment, which means they must focus on continuously improving their processes. Eliminating waste is then crucial to ensure process efficiency, being the main target of both lean thinking concept and Toyota Production System (TPS). Another important factor to consider is the technological growth, which is beneficial to pursue and combine with continuous improvement and waste elimination. Despite TPS, Lean, and Kaizen traditionally focusing on productive processes and related activities, their fundamental principles are applicable to any part of a company, meaning that processes with the potential for automatization also include back-office processes, which rely heavily on information flows and employee knowledge. This study integrates back-office process improvement with the implementation of technology, through the development of a project based on the Continuous improvement philosophy and A3 Thinking methodology, consisting on automatizing the documentation of standardized work analysis used by an automotive company.

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1. Introduction

The automotive sector has evolved exponentially over the past decades, becoming one of the most competitive industries today. Therefore, companies within this sector, besides the need to have strong production capabilities and high-quality products, also must actively and continuously improve their processes, making them more effective and efficient. Consequently, reducing or even eliminating waste is a crucial step to ensure process efficiency, being this the main target of the lean thinking concept which was based on Toyota Production System (TPS) [1; 2]. Moreover, another strategy that addresses the challenges posed by the competitive and volatile environment in which these companies operate, is based on technological growth [3]. Hence, it would bring companies many benefits if continuous improvement, waste elimination and technological growth were sought after and even combined. Both TPS, Lean and Kaizen have traditionally been focused on manufacturing or the shop floor processes and related activities, however, its basic principles are applicable to any part of a company [2]. The processes with the potential for automation include not only shop-floor processes but also back-office processes. [2], introduces the concept of a lean office, for which lean thinking is applied to the administrative areas of the company, achieving the benefits of lean production. The key distinction between lean production and lean office lies on their inherent processes. In lean production, the processes are highly visible due to the physical flows involved. Conversely, in a lean office, value-adding processes rely heavily on information flows and employee knowledge [4]. Regarding to the seek for technological growth, automatizing administrative tasks enhances employee productivity by reducing processing times, improving information sharing, and data access [5].

The aim of this study is to illustrate the development of an approach to back-office processes' improvement and its practical implementation in an automotive company. The study combines the concepts of back-office process improvement and waste reduction with the implementation of technology, in the development of a project to automatize the documentation of standardized work analysis in processes used by the company, that operates under the Toyota Production System (TPS) philosophy, where standard documentation described by the TPS is widely used. This project was based on the Continuous improvement philosophy and A3 Thinking methodology.

2. Theoretical Background

2.1. Standardized Work Documentation on TPS

Regarding standardized process documentation, in the context of Toyota Production System, as illustrated in Figure 1, standardized work is not defined by just one document, but rather by several that are interconnected to analyze the standardized work itself [6]. The documents share the common purpose of facilitating and enhancing work analysis and incorporate four of Ohno's principles for standardized work – Takt-Time, Work Sequence, Standard WIP, and Automation [1].

[6], describes each Toyota document as follows:

- Time Measurement Sheet – This document is used to understand the actual time required to complete operations, incorporating shop floor observations, considering variability. These studies are defined under the same conditions to ensure work replicability, with the goal of maintaining a consistent work sequence. This specific document divides operations into various work elements, such as starting points, operation times, and sequences, which together allow for the identification of waste points.
- Production Capacity Sheet – This document integrates the principle of automation, representing machine capabilities and setup times, manual operation times, and automatic work times.
- Line Balance Chart – This document is used to record the actual times to complete a set of operations, integrating the principles of Takt-Time and work sequence. It also provides cycle times for each workstation and helps visualize where balancing is needed.
- Work Chart – It is a document that analyzes activities associated with work, integrating Takt-Time, work sequence, and automation. This document illustrates workstation layouts, material flows, and equipment locations, allowing for the improvement of Takt-Time and work sequence automation times.
- Work Method Chart – Work sequence and Standard WIP are incorporated into this document, which assists operators in performing standardized work and providing training for standardized work execution.

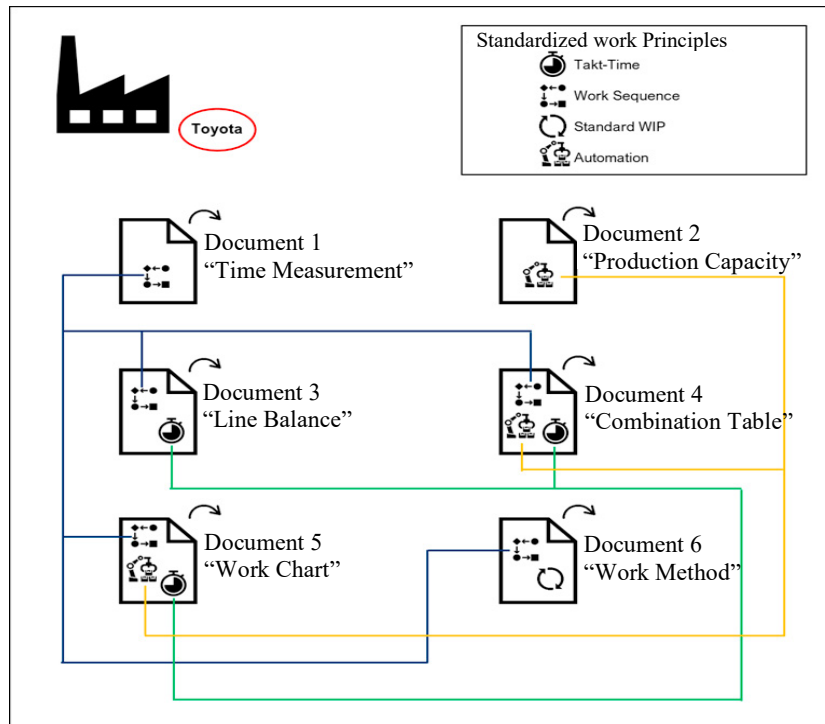


Figure 1 - Standardized Work Documents in Toyota (adapted from [6])

2.2. Process Improvement and Lean Wastes

The activity of improving a business process is formally known as Business Process Improvement (BPI), which is essentially a systematic approach aimed at helping an organization make significant changes to a specific business process [7]. The same author outlines a framework for this methodology composed of five phases: initialization, selection, design, implementation, and evaluation. The selection phase is considered crucial, as it involves selecting the activities or processes to be improved and the corresponding countermeasures to be implemented. Similar to the waste reduction perspective of Lean Production Systems, the purpose of the BPI methodology is to use resources more efficiently and effectively. This relationship between methodologies is further reinforced by [7], who applies a waste reduction framework to the selection phase of the BPI methodology. According to [8], Lean pure wastes which must be under attention for further reduction or elimination, can form the acronym DOWNTIME (Defects, Overproduction, Waiting, Non-utilized talent, Transportation, Inventory, Motion, Extra work/Excess Processing):

The role of waste reduction can be applied to a wide range of processes. In manufacturing industries, waste can be reduced or eliminated in the production process itself as well as in administrative office work, such as planning activities [7], production control activities, and activities related to information flows [9]. Regarding information flows, which are closely related to back-office processes, initially, there is a set of data that, in such context, is equivalent to raw material. Subsequently, data is stored, processed, organized, and finally disseminated, and within all those steps, waste can also be identified and subsequently eliminated. [9] draw a parallel between the formal and common definitions of the eight Lean wastes and their definition in the context of information flows, which is summarized in Table 1.

In the context of process improvement, [7] highlights the growing role of Information Technology (IT), which essentially comprises the capabilities provided to organizations by computers, software applications, and telecommunications, offering data, information, and knowledge. Thus, IT can and should serve as the foundation and support for process improvement activities, potentially innovating and simplifying them, automating their activities, and integrating them with others [7].

Table 1 - Lean wastes (DOWNTIME) on information flow (adapted from [9])

Waste Classification	Description from the standpoint of information flows
Defects	Inappropriate actions based on inaccurate information or lack of information, which may lead to inefficient decision-making.
Overproduction	Generation and maintenance of unnecessary and non-value-added information.
Waiting	Time and resources spent on identifying information that is not readily available, resulting in delays for the correct information.
Non-utilized Talent	Human resources required to work on non-optimized processes due to the existence of other wastes.
Transportation	Movement/flows of information that do not add value to the user receiving it.
Inventory	Databases and files that provide more information than necessary for the current moment.
Motion	Unnecessary movement to search for information due to inadequate organization and representation of information.
Extra Work	Activities carried out to compensate for the lack of information, which may include creating or seeking additional information.

Two studies regarding the practical implementation of lean principles were found. [10] explored the implementation of lean principles in a public sector office environment, applying some principles and techniques. Process mapping was used to identify waste in administrative processes, being each task analyzed to determine if it added value, which led to eliminating unnecessary steps and optimizing the flow of tasks. Also, the 5S methodology was implemented to improve workspace efficiency. By sorting, setting in order, shining (cleaning), standardizing, and sustaining discipline, teams organized office spaces and filing systems, making it easier to find documents and tools, thereby enhancing productivity. Moreover, processes were standardized across the departments to ensure consistency and reduce errors. This included creating standardized filing systems and procedures, which reduced time spent searching for documents and made workflows more predictable and efficient. On the other hand, [11] details the application of Kaizen events to improve a recruitment process within a manufacturing company, highlighting the effectiveness of Kaizen in improving administrative tasks, specifically recruiting co-ops for frontline supervisor positions. This study then emphasizes the potential of Kaizen events in non-production settings, showing their broader applicability across various business functions. The identification of only two relevant studies points to a significant gap in the literature on improving non-manufacturing processes, especially in terms of practical implementation.

3. Case Study

3.1. Contextualization

The implementation of standardized work documentation to define the current status of processes is part of the work culture within the company where this study took place. Therefore, in order to contextualize this case study, it is important to understand the documents that are used by the company, which are described as follows:

- **Standardized Work Recording Sheet (SWRS):** This document is filled out based on recordings made of activities at each workstation. It is important that these recordings capture the operators' hand movements. Each work element and its respective duration are registered in the document.
- **Work Step Sheet (WSS):** This sheet represents all operations and their execution times divided into different parts of the production sequence: manual work time, machine processing or waiting time, and travel time.
- **Standardized Work Combination Table (SWCT):** This document translates the sequence of work elements and their execution times and presents the interaction and combination of different parts of the production

sequence: manual work time, machine processing or waiting time, and travel time. This document helps identify waste, such as waiting times, overload, and excessive travel times.

- Yamazumi Chart (YC): This is a stacked bar chart where each bar represents an operation. It also includes a line representing the takt time. This allows for a visual understanding of operation balancing.
- Standardized Work Chart (SWC) or Moving Line: This is a spaghetti diagram showing the operator's movements in the production sequence, material location, and process layout.
- Process Yamazumi Chart or Weighted Average Cycle Time (WACT): This chart reflects relevant information for the process and consists of a bar chart representing the cycle time for each model in a mixed-model production environment, allowing calculation/visualization of a weighted average cycle time.

3.2. Problem Description and Methodology

The implementation of standardized work documentation requires filling out each document with information gathered on the shop-floor, this being a very exhaustive and long activity, also prone to errors. Accordingly, it was intended to improve the set of documents making the whole process of implementation more efficient.

In order to tackle this challenge, an improvement process was conducted following the A3 Methodology, in which one of the first steps was to get a grasp of the current situation and corresponding root-causes. To this end, a root cause analysis was conducted resorting to the Ishikawa Diagram and the 5 Whys. After identifying the main cause of the problem, aiming to solve it, the 5S methodology was implemented to define the main aspects that should be considered in the design of the new solution, for which Excel and VBA programming were used.

The methodology used to collect information and to develop and implement the solution was a hybridization of direct observation and recurring meetings with all process engineers which were involved in the activities of standardized work documentation, during a period of four months.

3.3. Problem Analysis, Solution Implementation and Results

Regarding the root cause analysis, Figure 2 shows the results found through the implementation of the Ishikawa Diagram. For the inefficient filling out process of standardized work documentation, three classes of causes were considered: Man, Method, and Machine. For the Man class, the main cause found was the occurrence of errors and high subjectivity throughout the documentation process, whereas for the Method class, it was noted that there was a need to replicate data and records between multiple documents. Finally, for the Machine class, which in this case is the company's shared server, the main issue was the long time that took to find a specific needed document. For each of this causes, the 5 Whys tool was applied, as shown in Figure 3, to identify the root causes, these being the lack of connection between various documents and the absence of automatization in filling them out and the lack of standard rules and procedures for the documentation process. Therefore, it became clear that the aim of the project would be to modify the standardized work documentation so that its filling and consequential application became more agile and better defined. The tools selected for conducting this analysis are widely used within process improvement activities in the context of the implementation of TPS [1][8].

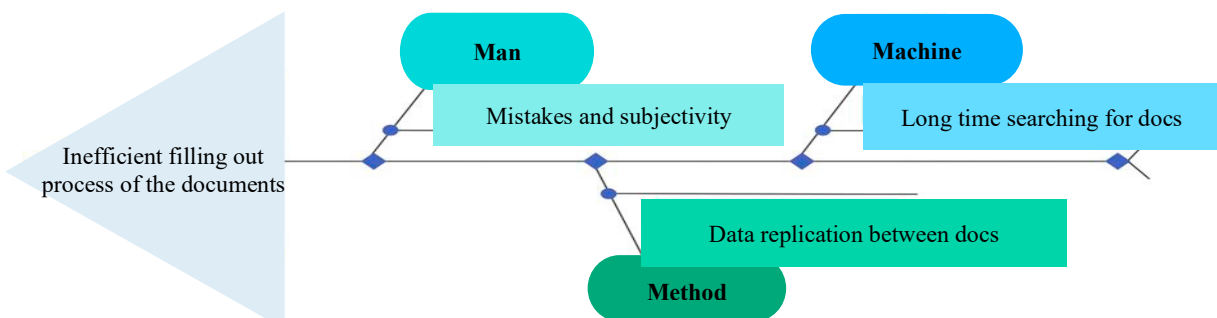


Figure 2 - Ishikawa Diagram for identifying the main causes

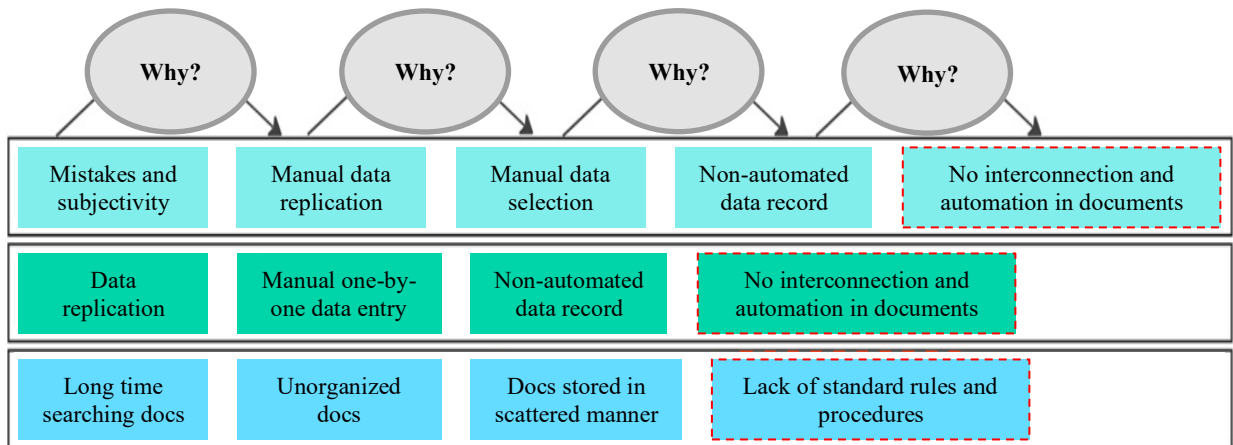


Figure 3 - 5 Whys for root cause analysis

3.4. Solution Implementation and Results

Having identified the root causes, the 5 S methodology was used to guide the design of a solution, as shown in Table 2. This solution revolved around the basic acknowledgment that the SWRS serves as the foundation for all the other documents, which meant that the design approach was to give to this document the role similar to a database. To this end, various tools and functionalities of Excel and VBA programming were used. Thus, the automation of documentation was implemented through the creation of dynamic and interconnected sheets and documents, automatic calculations, creation of macros, and also through the creation of dynamic reports and charts.

Table 2 - 5S Implementation for the design of the new documentation process

5 S	Modification and automation of documentation – 5 S Implementation
Seiri (Sort)	Eliminate unnecessary actions in the filling out process.
Seiton (Straighten)	Improve the organization system in shared folders within the company.
Seiso (Shine)	Improve the layout/design of the documents.
Seiketsu (Standardize)	Develop filling out procedures for the standardized process.
Shitshuke (Sustain)	Schedule training sessions and continuous communication to make this new approach an integral practice of the department.

Regarding the new set of documents developed, Figure 4 shows the automation differences between them, and the set of documents used before. The initial set of documents was highly characterized by a very manual process where most of the data registered in the documents was filled out manually. Contrarily, the new set of documents was mostly comprised by automatically filled out fields. Moreover, this modified set of documents provides more information than the previous one and also incorporates two more documents work analysis documents, adding even more information to work analysis. These new documents - Operations Yamazumi (OY) and Anomaly Analysis Report (AAR) - were also integrated with the automated documents described above. These new documents' aim is described as follows:

- OY: The purpose of OY is to analyze value-added, non-value-added, and waste in each macro-operation, aiming to identify macro-operations with higher waste indices for subsequent analyses focused on waste reduction.
- AAR: The purpose of the ARR, is to identify anomalies within the process, which are, by definition, activities which are not part of the standardized work and therefore should be eliminated. As such, this document registers identified anomalies and their corresponding times, as well as the operations to which they are associated. To classify the found anomalies, scores are assigned to them, regarding the frequency of occurrence and severity. Finally, an overall score is assigned to each anomaly, functioning as a warning to prioritize efforts to eliminate those anomalies. This score is generated using the Analytic Hierarchy Process (AHP) multicriteria decision-making method.

Furthermore, the way the initial set of documents was stored was also modified. Process analysis and subsequent filling out of the standardized work documentation are activities carried out for all stages of the production process and as such, each stage corresponds to a folder on the company's server. Before the implementation of this project, documentation for each stage of the process was stored in the corresponding folder without any specific criteria. However, after developing and implementing the new standardized work documentation, predefined standard rules and procedures were created regarding both the process of filling out each document and storing them in the company's server.

Docs. Documents used before Automation

Most of the SWRS had to be filled out manually.
Number of **manual** fields = 4
Number of **automatic** fields = 4

SWRS

Documents used after Automation

New fields were added to facilitate the connection with the other documents: Operation Number (manual) and Operation Type(automatic).
Number of **manual** fields = 4
Number of **automatic** fields = 8

Besides the graph, the whole document had to be filled out manually.
Number of **manual** fields = 7
Number of **automatic** fields = 1

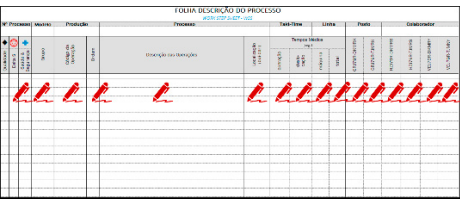
SWCT

The whole document was automatically filled out.
Number of **manual** fields = 0
Number of **automatic** fields = 11

Docs. Documents used before Automation

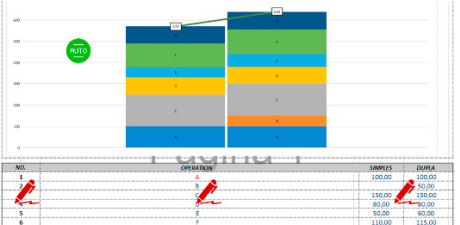
The whole document had to be filled out manually.
Number of **manual** fields = 16
Number of **automatic** fields = 0

WSS



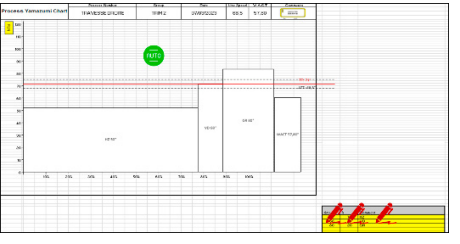
It was necessary to manually enter the data to generate the yamazumi chart.
Number of **manual** fields = 3
Number of **automatic** fields = 1

YC



It was necessary to manually enter the data to generate the yamazumi chart.
Number of **manual** fields = 3
Number of **automatic** fields = 1

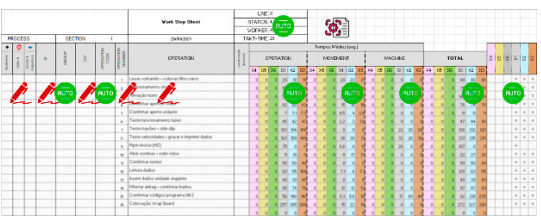
WACT



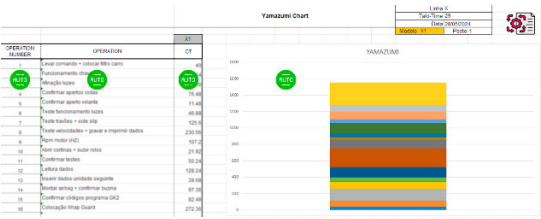
OY

Documents used after Automation

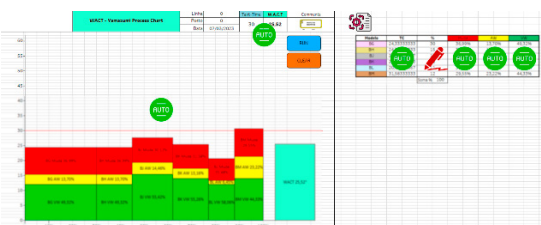
Most of the document was automatically filled out.
Number of **manual** fields = 4
Number of **automatic** fields = 8



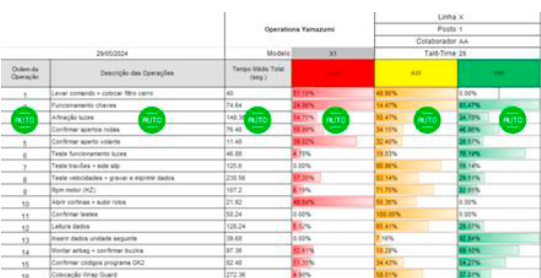
The whole document was automatically filled out.
Number of **manual** fields = 0
Number of **automatic** fields = 4



Most of the document was automatically filled out. Also, the modifications made to this document allowed to know more information such as the percentages of MUDA, AW, and VW activities for each model produced in the company.
Number of **manual** fields = 1
Number of **automatic** fields = 6



All information is automatically generated.
Number of **manual** fields = 0
Number of **automatic** fields = 6



Docs.	Documents used before Automation	Documents used after Automation
AAR		Some fields are automatically generated and others, which refer to new information that is not provided by any of the previous documents, need to be manually filled out. Number of manual fields = 6 Number of automatic fields = 2 

Figure 4 - Comparison between initial and new standardized work documents

After developing the new set of documents and implementing them within the context of the company's process, it was possible to evaluate the impact of the project. Regarding the filling out time, for the initial set of documents it was necessary approximately 25 hours to complete the whole standardized work documentation. Conversely, for the developed documentation it took about 15 hours to complete it, which can be translated by a 40% filling out time reduction.

The impact of this project can also be evaluated according from the perspective of Lean waste reduction. Having applied all 5S during the development of this project, it was possible to eliminate or reduce four Lean Wastes, as shown by Table 3: defects, underutilization of talent, transportation and overprocessing. For instance, regarding the overprocessing waste, by automatizing most of the documents' fields, the need to replicate data entry between documents was eliminated, which impacted the time spent on the documentation process. Thus, by reducing time spent on filling out documents, employees were more available to engage in the true analysis of the shop-floor processes and more creative tasks.

Table 3 - Lean Wastes' elimination through documentation's automation

Lean Wastes Types	Modification and Automation of Documentation – Waste Elimination
Defects	Through the new automatized solution, errors in data transcription and information transfer from document to document were prevented.
Underutilization of talent	The new solution reduced the time spent on filling out documents, freeing up employees to engage in more creative tasks.
Transportation	With the new system for organizing documentation in shared folders and the creation of a document access menu, it became easier to find and quickly visualize the desired documentation.
Overprocessing	Through the new automatized solution, the need to replicate the entry of data and information from document to document was eliminated.

4. Conclusions

Typically, continuous improvement is associated with shop-floor operations. However, this study demonstrates that it can be highly effective for back-office processes. Through the automation of the standardized work documentation, significant efficiency gains were achieved, reducing documentation completion time by 40% whilst minimizing errors. The automation, facilitated by tools such as Excel and VBA programming, integrated different documentation sheets, automating data flows between them. Furthermore, the project emphasized how automation

can extend the principles of Lean and the Toyota Production System (TPS) to back-office functions, streamlining tasks traditionally handled manually. Automating these processes not only substantially reduced overprocessing and errors, which are key Lean wastes, but also minimized transportation waste in the form of information retrieval. It also tackled the underutilization of talent by allowing workers to focus on more value-adding tasks like the effective shop-floor process analysis engage, these being in more creative and analytical activities.

This study illustrates the broad applicability of continuous improvement approaches and tools, with automation acting as a conveyor for more efficient, streamlined processes across all business functions. It reinforces the notion that back-office processes, often overlooked, can benefit greatly from Lean principles when combined with automation technologies, ultimately driving more effective business operations. This project serves as an example of how the used tools can extend the impact of Lean methodologies beyond the production floor into the back-office heart of organizations.

Acknowledgements

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References

- [1] Ohno, T. (1988). *Toyota Production System: Beyond Large-Scale Production*. Productivity, Inc.
- [2] das Chagas Santos, J., Freitag, A.E.B., Quelhas, O.L.G. (2020). Lean Office and Digital Transformation: A Case Study in a Services Company. In: Leiras, A., González-Calderón, C., de Brito Junior, I., Villa, S., Yoshizaki, H. (eds) *Operations Management for Social Good. POMS 2018. Springer Proceedings in Business and Economics*. Springer, Cham. https://doi.org/10.1007/978-3-030-23816-2_92
- [3] Guskova, M. F., & Nepopushcheva, A. O. (2019). The Management and Automation of Manufacturing Processes. *Proceedings of the 2019 IEEE International Conference Quality Management, Transport and Information Security, Information Technologies IT and QM and IS 2019*, 8–10. <https://doi.org/10.1109/ITQMIS.2019.8928326>.
- [4] McManus, H.: *Product Development Value Stream Analysis and Mapping Manual (PDVMS)*. The Lean Aerospace Initiative: Massachusetts Institute of Technology (2005)
- [5] Noumani, R. A. Al, & Syedi, R. T. (2020). Human Resource Automation: Benefits and Challenges for Organizations. *Singidunum University International Scientific Conference*, 161–164. <https://doi.org/10.15308/finiz-2020-161-164>.
- [6] Marksberry, P., Rammohan, R., Vu, D., Marksberry, P., Rammohan, R., & Vu, D. (2011). A systems study on standardised work: a Toyota perspective. *Int. J. Productivity and Quality Management*, 7(3), 287–303.
- [7] Theuerkauf, J. “Jet” (2021, September 8). Four Ways To Approach Process Automation. <https://www.forbes.com/sites/forbesbusinessdevelopmentcouncil/2021/09/08/four-ways-to-approach-process-automation/?sh=727d2666220f>
- [8] Liker, J. K. (2004). *The Toyota way: 14 management principles from the world’s greatest manufacturer*. McGraw-Hill.
- [9] Lopes, R. B., Teixeira, L., & Ferreira, C. (2019). Lean Thinking across the Company: Successful Cases in the Manufacturing Industry. In F. J. G. Silva & L. C. P. Ferreira (Eds.), *Lean Manufacturing: Implementation, Opportunities and Challenges* (pp. 1–30). Nova Science.
- [10] Monteiro, M. F. J. R., Pacheco, C. C. L., Dinis-Carvalho, J., & Paiva, F. C. (2015). Implementing lean office: A successful case in public sector. *FME Transactions*, 43(4), 303–310. <https://doi.org/10.5937/fmet1504303M>
- [11] Montabon, F. (2005). Using kaizen events for back office processes: The recruitment of frontline supervisor co-ops. *Total Quality Management and Business Excellence*, 16(10), 1139–1147. <https://doi.org/10.1080/14783360500235876>